SOFTWARE REQUIREMENTS SPECIFICATION · SALES ANALYTICS

Sales Analytics — SQL Analyst Requirements

End-to-end sales performance, customer, product and channel analytics for a multi-tier automobile spare parts distribution business — wholesale dealer, retail counter and online channels, from order through fulfilment.

DOCUMENT ID	**SRS-SALES-2026-01**	PREPARED BY	**Sales Analytics & Business Intelligence Team**
VERSION	**1.0**	DOCUMENT OWNER	**Director of Sales Operations**
STATUS	**Draft — Issued for Review**	DISTRIBUTION	**SQL Analyst — Sales Analytics**
CLASSIFICATION	**Internal — Confidential**	ISSUE DATE	**11 Sep 2026**

REVISION HISTORY

Version	Date	Author	Summary of changes
0.1	26 Aug 2026	Sales Analytics & Business Intelligence Team	Draft outline — scope and data-source audit against `auto_parts_analytics`.
1.0	11 Sep 2026	Sales Analytics & Business Intelligence Team	Baseline issued — sales-domain functional requirements, data dictionary, acceptance criteria.

TABLE OF CONTENTS

1.0 Introduction
2.0 Business Context
3.0 Data Source Overview
4.0 Functional Requirements
5.0 Non-Functional Requirements
6.0 Deliverables & Submission Standard
7.0 Assumptions & Constraints
8.0 Out of Scope
9.0 Approval & Sign-off

1.0 Introduction

1.1 Purpose

This document specifies the business and functional requirements for the Sales Analytics function, to be delivered entirely in SQL against the `auto_parts_analytics` PostgreSQL database. It is issued by Sales Analytics & Business Intelligence to the SQL Analyst(s) responsible for delivery. It defines what the business needs answered, the data available to answer it, and the acceptance bar a deliverable must clear — not how any individual query should be written.

1.2 Scope

In scope:

Revenue and sales trend analysis, sales channel and order-funnel performance, customer and dealer analytics, product and category sales performance, promotions and returns impact on sales, sales representative performance, and a consolidated executive sales KPI layer — across four full fiscal years (2022–2025) of transactional history.

Every requirement in Section 4 must be satisfiable using only SQL: standard queries, CTEs, window functions, and views / materialized views. No procedural extension (PL/pgSQL), external ETL, spreadsheet, or BI tool may carry analytical logic that SQL itself is capable of expressing.

Out of scope:

Inventory, finance / receivables and marketing-campaign analytics are separate domains and are explicitly out of scope — see Section 8.

1.3 Intended Audience

The SQL Analyst(s) assigned to Sales Analytics; Sales Operations stakeholders reviewing deliverables against the Section 4 acceptance criteria; and the Analytics Engineering function maintaining the underlying data platform.

1.4 Definitions, Acronyms & Abbreviations

Term	Definition
`SRS`	Software Requirements Specification — this document.
`KPI`	Key Performance Indicator — a metric the business tracks on a recurring cadence.
`RFM`	Recency, Frequency, Monetary — a customer segmentation model.
`CLV`	Customer Lifetime Value.
`AOV`	Average Order Value.
`OEM`	Original Equipment Manufacturer (vs. aftermarket part).
`CTE`	Common Table Expression — a `WITH` clause.
`YoY / MoM`	Year-over-Year / Month-over-Month.
`MNC`	Multinational Corporation — the calibre of analytical rigour this document is written to.

1.5 Reference Documents

- `sql/schema.sql` — authoritative DDL for all 32 tables (source of truth; supersedes the summaries in Section 3 if they ever diverge).
- `README.md` — environment setup and connection instructions.

2.0 Business Context

2.1 Business Background

The business is a multi-tier automobile spare-parts distributor. Product reaches market through three concurrent sales channels out of 15 regional warehouses — wholesale to Dealers (tiered Platinum / Gold / Silver / Bronze, on credit terms), direct to Retail counter customers, and direct-to-consumer via an Online storefront. Every order carries a downstream shipment and, where applicable, a return record, so the data supports the full order lifecycle, not just point-in-time revenue totals.

CHANNEL	ORDER LIFECYCLE	FULFILMENT
Dealer (800) Retail (50K) Online	Sales Order → Line ↓ Shipment ↓ Return (if any)	15 Warehouses

Procurement / inventory, receivables, and marketing / digital-campaign data exist in the same platform but sit outside this document's scope (Section 8) — they are governed by their own SRS documents within the same series.

2.2 Business Objectives

- Understand where revenue is being made and lost — by channel, region, dealer tier, product category and time period.
- Know the customer and dealer base well enough to act on it: who is valuable, who is churning, which acquisition channel actually pays back.
- Understand what is driving and dragging the order funnel — cancellations, returns, and fulfilment friction.
- Know which products and categories to double down on, and which to retire.
- Quantify what promotional discounting is actually buying the business in incremental sales.
- Know which sales channels, dealers and representatives are performing, and which need intervention.
- Produce a small set of trustworthy SQL views a Sales Operations leadership team could actually be handed.

2.3 Dataset at a Glance

2022–2025	600,000	3 channels	18 tables
4 fiscal years, daily grain	Sales orders	Dealer · Retail · Online	In Sales Analytics scope

3.0 Data Source Overview

This section is the analyst's map of the terrain before Section 4 hands out the actual work — what system holds the data, how much of it there is, and how the tables connect.

3.1 System of Record

PostgreSQL database `auto_parts_analytics`, single schema (`public`), 32 tables in total, fully normalised with declared primary / foreign keys and check constraints — verified against the live instance, not assumed from the DDL alone. No nulls or orphaned rows were found on the core order chain during data-source validation (0 `sales_order_lines` without a parent order). This SRS scopes the 18 tables that sit directly in the sales value chain, listed below; the remainder support procurement / inventory, finance and marketing domains and are out of scope (Section 8).

3.2 Data Domain Inventory — In Scope

Row counts as verified against the live database on 08 Aug 2026.

Domain	Key tables	Approx. rows	Grain
Sales	`sales_orders`, `sales_order_lines`, `shipments`, `returns`, `return_lines`	600,000 orders	Order › line
Customer & Dealer Master	`retail_customers`, `dealers`, `employees`	50,000 customers · 800 dealers · 150 employees	Entity master
Product Master	`parts`, `part_categories`, `manufacturers`, `vehicle_makes`, `vehicle_models`, `vehicle_model_years`, `part_vehicle_compatibility`	25,000 parts	Entity master + compatibility
Pricing	`price_history`	44,972 rows	Part × effective date
Promotions	`promotions`	500 promo codes	Referenced directly by sales orders
Location Reference	`warehouses`	15 warehouses	Fulfilment / region dimension

3.3 Table Inventory — Sales Core

The tables every functional requirement in Section 4 ultimately traces back to. Full column-level detail is authoritative in `sql/schema.sql`; this is the subset an analyst touches daily.

Table	Description	Key columns
`sales_orders`	One row per order; channel, dealer / customer, status, money totals.	`order_id · order_date · channel · order_status · promotion_id · total_amount`
`sales_order_lines`	One row per part per order.	`order_id · part_id · quantity · unit_price · line_total`
`shipments`	Fulfilment status and dates per order.	`order_id · ship_date · delivery_date · shipment_status`
`returns / return_lines`	Post-sale returns and per-line refund amounts.	`order_id · reason · status · refund_amount`
`parts / price_history`	Product master and its cost / list price over time.	`cost_price · list_price · category_id · manufacturer_id · effective_date`
`part_categories / manufacturers`	2-level category hierarchy; OEM vs. aftermarket flag.	`parent_category_id · category_level · is_oem`
`dealers / retail_customers`	The two customer masters feeding Dealer, Retail and Online orders.	`dealer_tier · credit_limit · region · signup_date · acquisition_channel`
`employees`	Sales representatives linked to orders and warehouses.	`employee_id · role · region · warehouse_id`
`promotions`	Discount codes referenced directly by `sales_orders.promotion_id`.	`promo_code · discount_type · discount_value`
`vehicle_makes / models / model_years / part_vehicle_compatibility`	Cross-sell dimension — which parts fit which vehicles.	`make_id · model_id · model_year · part_id`

4.0 Functional Requirements

Seven modules, ordered as a Sales Analytics function would tackle them — from top-line revenue down to a consolidated executive view layer. Each module states the objective, the business questions it must answer, the tables it draws on, and the acceptance bar. The acceptance criteria in each module define the minimum bar for sign-off. Each module carries a priority of MUST or SHOULD.

ID	Module	Priority
FR-1	Revenue & Sales Trend Analysis	**MUST**
FR-2	Sales Channel & Order Funnel Performance	**MUST**
FR-3	Customer & Dealer Sales Analytics	**MUST**
FR-4	Product & Category Sales Performance	**MUST**

ID	Module	Priority
FR-5	Promotions & Returns Impact on Sales	SHOULD
FR-6	Sales Performance — Representatives, Regions & Warehouses	SHOULD
FR-7	Executive Sales KPI View Layer	MUST

FR-1 Revenue & Sales Trend Analysis — MUST

Establish the top-line revenue baseline and show how it moves across channel, region and time — the number every other module gets checked against.

BUSINESS QUESTIONS

- Total recognised revenue by year and month, with MoM and YoY growth.
- Average order value and order volume by channel — Dealer vs. Retail vs. Online.
- Revenue leaders and laggards by region.
- Revenue lost to cancellations, and discount rate as a % of gross.

PRIMARY TABLES

`sales_orders` · `sales_order_lines` · `dealers` · `warehouses` · `retail_customers`

ACCEPTANCE CRITERIA

- ✓ MoM / YoY growth computed with window functions (`LAG`), not self-joins.
- ✓ Cancelled orders excluded from revenue and separately quantified as leakage.
- ✓ Channel × region matrix produced.
- ✓ Discount expressed as a % of gross, not only an absolute figure.

`Window Functions` `Date Bucketing` `ROLLUP`

FR-2 Sales Channel & Order Funnel Performance — MUST

Compare how Dealer, Retail and Online actually perform, and trace the order lifecycle from placement to delivery or cancellation.

BUSINESS QUESTIONS

- Order-status funnel — what share of orders reach Confirmed, Shipped, and Delivered, and what share end in Cancelled?
- Average order-to-ship and order-to-delivery cycle time by channel, from shipments.
- Shipment status distribution (Delayed / Returned) and its relationship to channel and region.
- Channel-wise order conversion & cancellation rate.

PRIMARY TABLES

`sales_orders` · `shipments`

ACCEPTANCE CRITERIA

- ✓ Funnel stage transition rates computed relative to total orders entering the pipeline, not just final-status counts.
- ✓ Cycle time calculated per order and aggregated (avg / median), not derived from pre-aggregated numbers.
- ✓ Cancelled orders analysed as a distinct funnel exit, never merged into the Delivered denominator.

`CASE / FILTER` `Funnel` `Date-Diff`

FR-3 Customer & Dealer Sales Analytics

MUST

Turn the two customer masters — dealers and retail customers — into a segmentation the business can actually act on.

BUSINESS QUESTIONS

- RFM segments for retail customers — who sits in the top decile?
- An approximate CLV per customer and per segment.
- Cohort retention by signup month — what share of a cohort is still ordering N months on?
- How does dealer tier performance differ in terms of average credit limit, average delivered revenue per dealer, and credit utilisation?
- Which `acquisition_channel` produces the highest-value customers?

PRIMARY TABLES

`retail_customers` · `sales_orders` · `dealers`

ACCEPTANCE CRITERIA

- ✓ RFM scored with quantile logic (`NTILE`), not arbitrary manual cutoffs.
- ✓ Cohort retention is reported by signup month, months since signup, and retention percentage.
- ✓ Dealer tier analysis reconciles `credit_limit` against realised order value.

`NTILE` `Cohort Matrix` `Self-Join`

Segment	R Score	F Score	M Score
Champion	5	5	5
Loyal	4–5	3–5	Any
Risk	2–5	1–2	Any
Lost	1	1	Any
Intermediate	Everything else		

FR-4 Product & Category Sales Performance MUST

Identify what sells, what's actually profitable once cost is accounted for, and how pricing decisions have moved either.

BUSINESS QUESTIONS

- Analyse revenue and gross margin by category and sub-category using the part category hierarchy.
- Rank parts by revenue and calculate their cumulative contribution to total revenue, showing the top 100 parts by revenue.
- Identify the number and percentage of parts that collectively generate 80% of total revenue.
- Compare OEM vs. non-OEM manufacturers by distinct parts sold, revenue, gross margin, and margin percentage.
- Analyse whether list price changes are associated with shifts in unit volume by comparing sales volume 30 days before and after each price change.
- Identify the top 25 vehicle makes and models by revenue generated from compatible parts.

PRIMARY TABLES

`parts` · `part_categories` · `manufacturers` · `sales_order_lines` · `price_history` · `part_vehicle_compatibility` · `vehicle_models`

ACCEPTANCE CRITERIA

- ✓ Category relationships resolved dynamically through a recursive CTE.
- ✓ Pareto analysis uses a running cumulative-percentage window function.
- ✓ Margin uses the part's `cost_price` at time of sale, not just current list price.

Recursive CTE **Cumulative %** **Self-Join**

FR-5 Promotions & Returns Impact on Sales SHOULD

Quantify what promotional discounting is actually buying the business in incremental sales, and how much return activity is eroding net revenue.

BUSINESS QUESTIONS

- Promo-code redemption rate and average discount depth (`discount_amount` subtotal) by promotion.
- Does promoted order value / volume outperform non-promoted, net of the discount given?
- Return rate by category and channel, and its dollar impact on net sales (`refund_amount`).
- Most common return reasons, and which products or channels concentrate them.

PRIMARY TABLES

`sales_orders` · `sales_order_lines` · `promotions` · `returns` · `return_lines`

ACCEPTANCE CRITERIA

- ✓ Promo lift measured net of `discount_amount`, not gross order value.
- ✓ Return rate and refund impact reported at both channel and category grain.
- ✓ Return-reason distribution reported as a ranked breakdown.

CASE / FILTER **Attribution** **Ratio Metrics**

FR-6 Sales Performance — Representatives, Regions & Warehouses SHOULD

Rank internal sales performance by employee and warehouse to identify who is driving — or dragging on — revenue.

BUSINESS QUESTIONS

- List the top 10 sales representatives ranked by realised revenue.
- Rank the top 3 regions based on average order value.
- List all warehouses ranked by utilisation rate (highest first).
- Find and list warehouses by return rate.

PRIMARY TABLES

`sales_orders` · `employees` · `warehouses`

ACCEPTANCE CRITERIA

✓ Tied reps must share a rank, with the earlier employee listed first.

`RANK / DENSE_RANK` `Conditional Aggregation`

FR-7 Executive Sales KPI View Layer MUST

Consolidate FR-1 through FR-6 into a small, reusable set of SQL views a Sales Operations leadership audience could be handed directly — the deliverable the rest of this document builds toward.

REQUIRED VIEWS

- Monthly Sales Performance Growth
- Customer Segments
- Dealers Performance by Credit Utilisation
- Product & Category Performance
- Sales Order Funnel Performance
- Sales Representatives Performance
- Regional & Warehouse Performance

SOURCE MODULES

`FR-1` · `FR-2` · `FR-3` · `FR-4` · `FR-5` · `FR-6`

ACCEPTANCE CRITERIA

✓ Each view is independently queryable with a plain `SELECT *` and returns a documented grain.

✓ Views reuse FR-1 … FR-6 logic rather than re-deriving it differently.

`CREATE VIEW`

5.0 Non-Functional Requirements

5.1 SQL-Only Constraint

Every requirement in Section 4 — aggregation, ranking, segmentation, cohorting, funnel drop-off — must be expressed in standard SQL: `SELECT` / CTE / window-function pipelines and views. No PL/pgSQL procedures, no Python / pandas, no spreadsheet or BI-tool calculation may carry logic SQL can express natively. A thin, optional visualisation layer reading from a finished view is permitted but is never where analytical logic lives.

5.2 Performance

The order-line fact table alone is 1.81M rows across four years — large enough to punish careless queries. Queries must filter on indexed columns (`order_date`, `customer_id`, `dealer_id` are already indexed — see `sql/schema.sql`), avoid `SELECT *` against fact tables, and bound date ranges explicitly rather than scanning all four years by default.

5.3 Code & Documentation Standard

- One `.sql` file per business-question task within each functional requirement (e.g. an FR with four business questions becomes four files), named accordingly.
- Each file opens with a header comment: objective, business question(s) answered, tables used.
- Multi-step logic is built as named CTEs, one transformation per step — not a single unreadable nested query.
- Reusable results (FR-7) are materialised as SQL views, not copy-pasted queries.

5.4 Environment

PostgreSQL 14 or later. Window functions, `FILTER` clauses, `GENERATE_SERIES` for date spines, and recursive CTEs (for the two-level category hierarchy) are all expected to be used where they are the right tool — this document assumes and rewards fluency with them, not avoidance of them.

6.0 Deliverables & Submission Standard

- One SQL script per business-question task within each functional requirement in Section 4 (`fr1_revenue_trend.sql` … `fr7_executive_views.sql` naming, with a suffix per task) — not one script per module.
- A view layer for FR-7 — a small number of named, queryable views Sales Operations leadership could be handed directly.
- A one-page findings note per module (Markdown, not a slide deck): the two or three things the data actually shows, in business language, each backed by the query that produced it.
- No modification of the source data itself — the schema in `sql/schema.sql` and the data in `auto_parts_analytics` are fixed inputs, not part of the deliverable.

7.0 Assumptions & Constraints

Topic	Assumption
Revenue recognition	Unless a requirement states otherwise, recognise revenue on `order_status = 'Delivered'`; exclude Cancelled orders; treat Pending / Confirmed / Shipped as pipeline, not booked revenue. A different definition may be used only if stated explicitly alongside the result.
Currency	Single currency — no foreign-exchange conversion is required or modelled.
Time zone	No time-zone handling — all timestamps are server-local; no multi-region time normalisation is required.
Data privacy	No PII masking or access-control layer is required for this engagement.

8.0 Out of Scope

The areas below are excluded from this engagement. Promotion redemption and its direct impact on sales (FR-5) remain in scope, since `promotion_id` sits directly on `sales_orders`.

Excluded area	Covers	Governed by
Procurement & inventory analytics	Purchase orders, stock ledger, supplier reliability, stock transfers.	Separate Supply Chain Analytics SRS
Finance — receivables & payables	DSO, aging, collections, AP mirror.	Separate Finance Analytics SRS
Marketing campaign ROI & digital funnel analytics	Campaign performance, ad spend, web session / clickstream funnels.	Separate Marketing Analytics SRS
Upstream data pipelines	Building or modifying them — this database is a fixed input to the engagement.	—
Front-end dashboard / BI-tool construction	Power BI, Tableau, Looker — a stretch option, never a requirement.	—
Predictive modelling	Forecasting, or any machine-learning technique.	—
Infrastructure & administration	Infrastructure, access control, or database administration.	—

9.0 Approval & Sign-off

This SRS is baselined at Version 1.0 and issued for the Sales Analytics engagement. Material changes to Section 4 scope require a version increment and an entry in the Revision History on the cover page.

Role	Name / function	Signature	Date
Prepared by	Sales Analytics & Business Intelligence Team		
Reviewed by	Manager, Sales Analytics — pending		
Approved by	Director of Sales Operations		